EDUHUB

Database Design & Data Modeling

Entity-Relationship Diagram

1. Introduction

This document is the unified database design of EduHub, an online learning platform built with ASP.NET Core MVC, Entity Framework Core, SQL Server, ASP.NET Core Identity and Stripe. It presents the ER diagram and relationships, the logical and physical schema of all 23 tables, the keys, indexes and constraints, the business rules the schema supports, and the normalization analysis.

1.1 Approved Scope

- EduHub delivers recorded video lessons only, organized in sections and lessons. No live-session features are part of the scope.
- Paid access is unlocked only by verified Stripe payments. Card numbers and security codes are never stored.
- Progress is tracked per lesson; section quizzes and a final exam govern course completion and certificate issuance.
- The model contains exactly the 23 entities of the approved ERD; no additional features are modeled.

1.2 Conventions

Term	Meaning
PK / FK / UQ	Primary key, foreign key, unique constraint. “PK, FK” is both. “→ Table” names the referenced table.
Cardinality	1 = exactly one, 0..1 = zero or one, 0..N = zero or many, 1..N = one or many.
Assumed	Data types, lengths, defaults, enumerated values, checks, indexes and delete rules are proposed implementation assumptions, to be finalized in the EF Core Fluent API configuration.
ProfileFields	Extended student/instructor profile data stored as JSON text in one column (assumed).

2. Entity-Relationship Diagram

2.1 Complete ERD

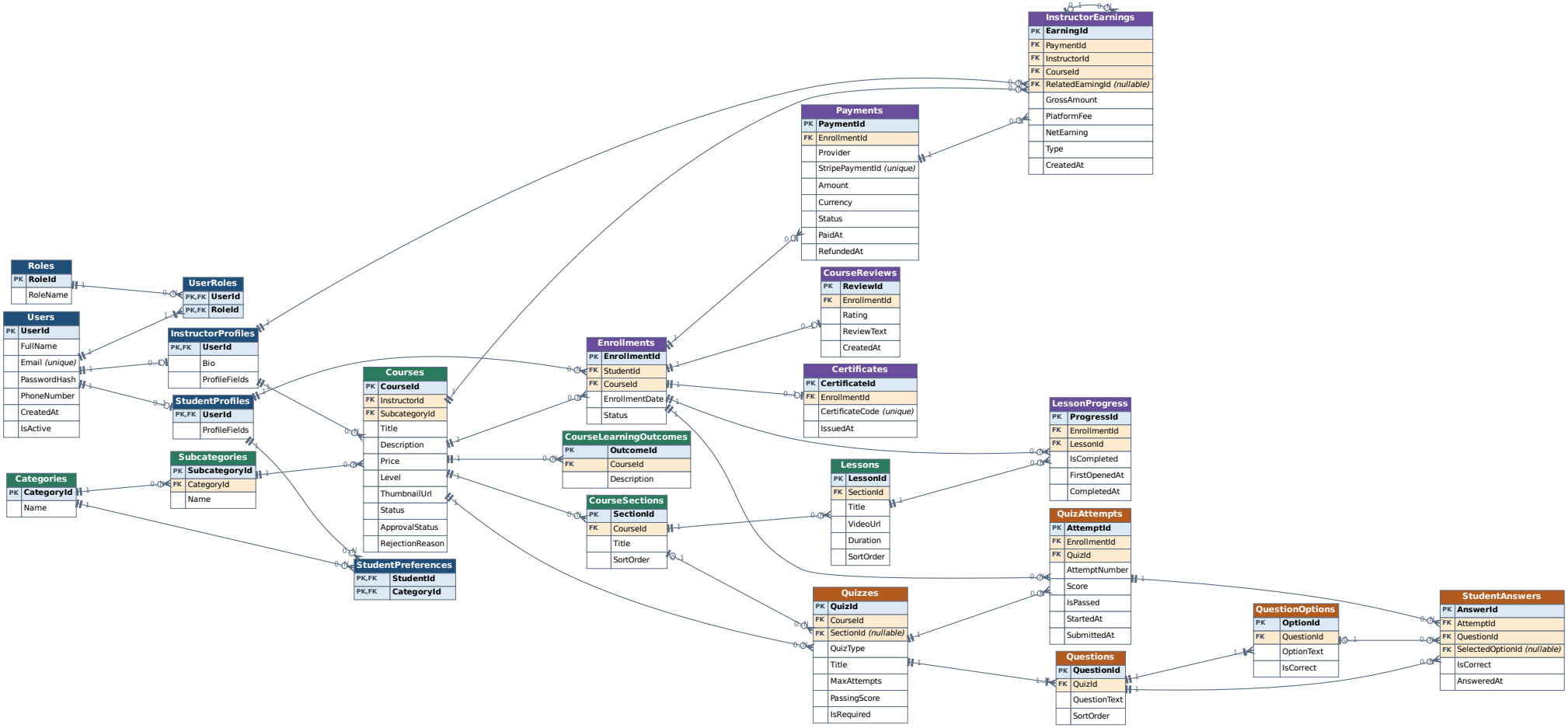
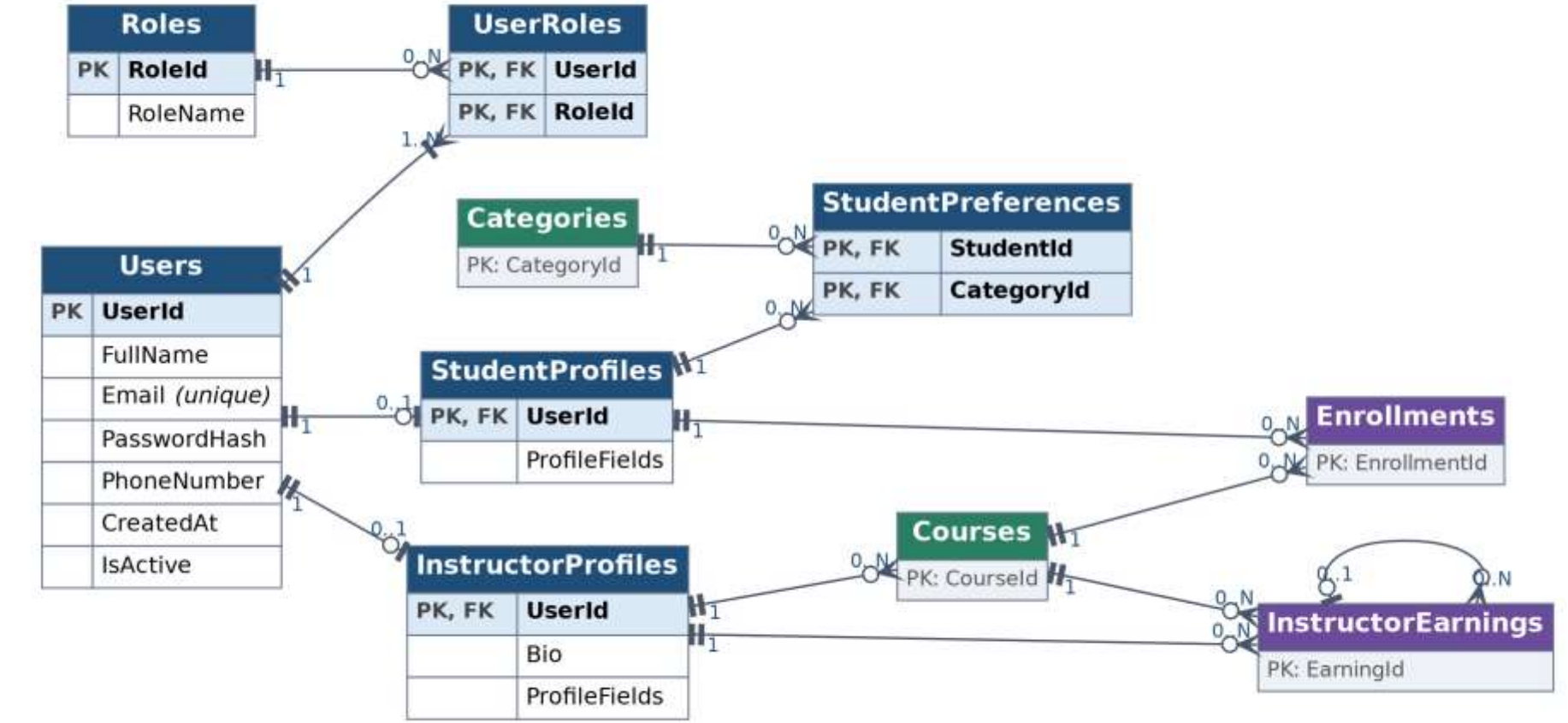


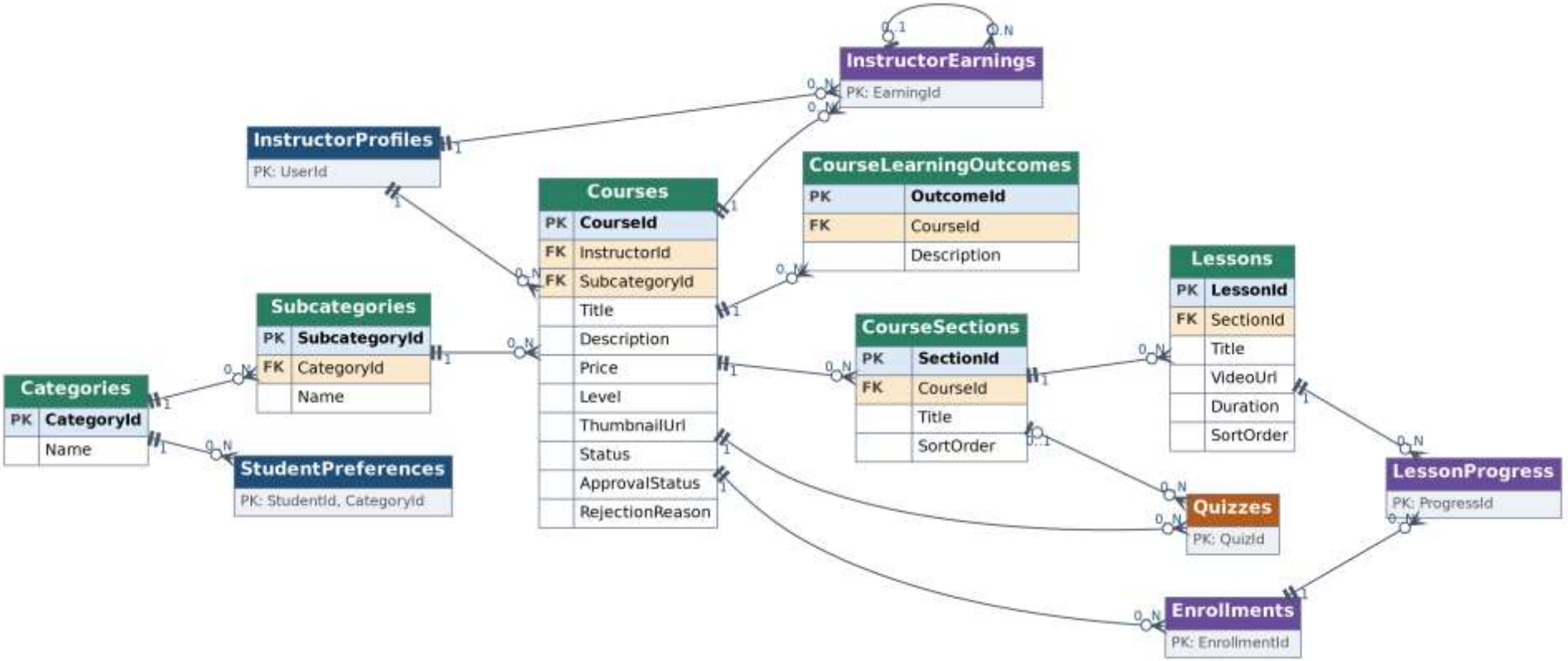
Figure 1 — Header color = subject area; light-blue rows = PK, light-orange rows = FK; line-end labels = cardinality; italics = unique / nullable.

2.2 Users & Access



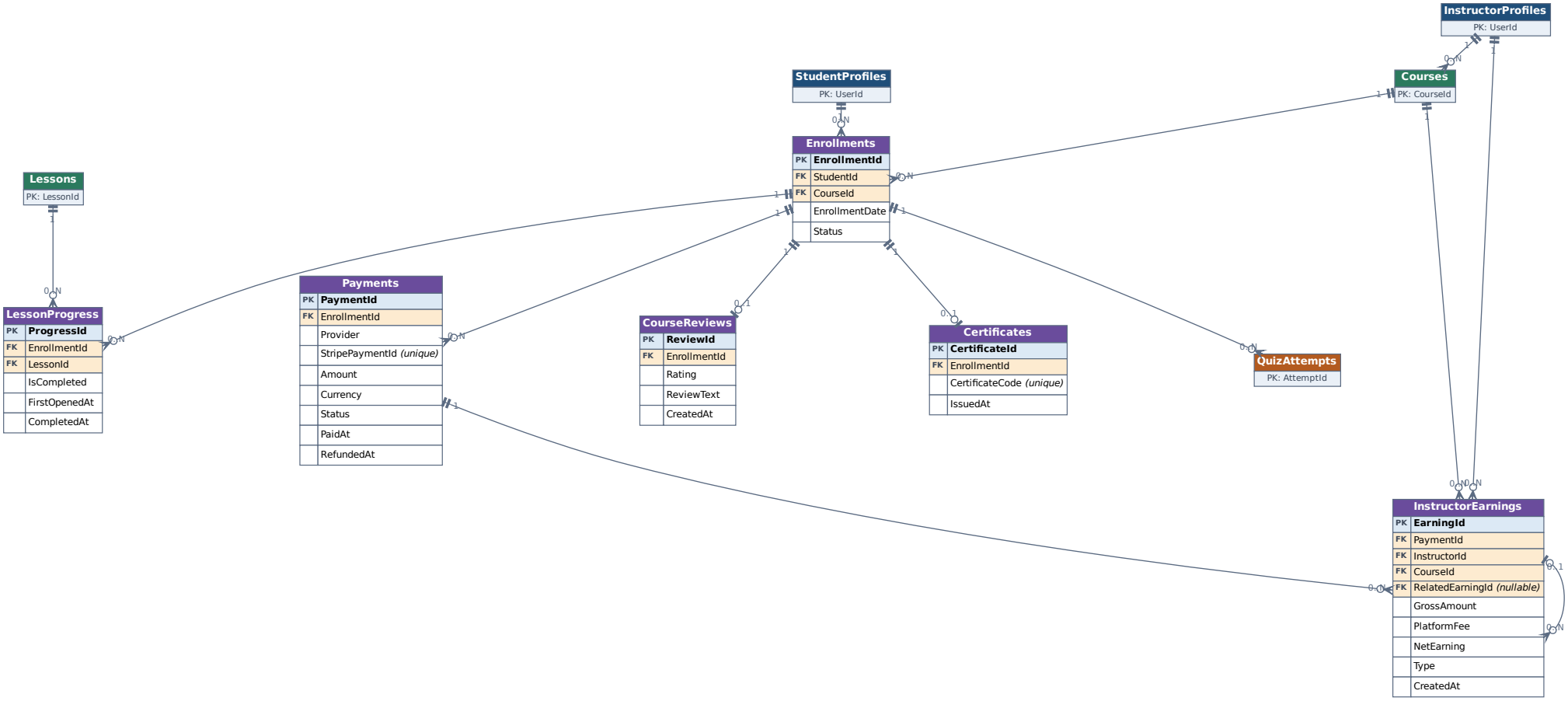
Users hold roles through UserRoles and may have a student profile, an instructor profile, or both. Students record preferred categories in StudentPreferences.

2.3 Course Catalog & Content



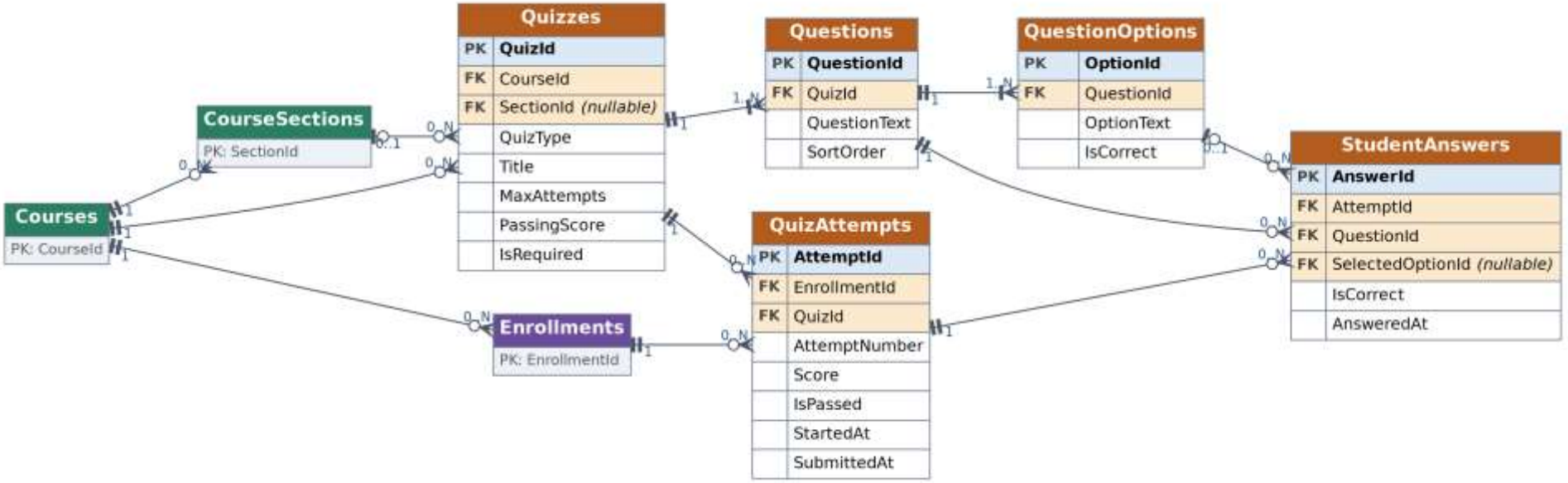
Categories contain subcategories; each course belongs to one subcategory and one instructor and is organized into sections, lessons and learning outcomes.

2.4 Enrollment, Payments & Earnings



An enrollment links a student to a course and anchors lesson progress, payments, reviews, certificates and quiz attempts. Each payment produces instructor earning entries.

2.5 Assessment & Quizzes



A quiz (section-level, or course-level / final exam when SectionId is NULL) contains questions and options. Each attempt stores one answer per question.

2.6 Subject Areas

Subject area	Purpose	Tables
Identity and Profiles	Accounts, roles, role assignment, the optional student and instructor profiles, and student category preferences.	Roles, Users, UserRoles, StudentProfiles, InstructorProfiles, StudentPreferences
Course Taxonomy and Catalog	Categories, subcategories, courses (pricing, publication status, approval status) and learning outcomes.	Categories, Subcategories, Courses, CourseLearningOutcomes
Course Structure	The internal structure of a course: ordered sections containing ordered recorded video lessons.	CourseSections, Lessons
Enrollment and Finance	Student enrollments, Stripe payments, and the instructor earnings ledger including refund reversals.	Enrollments, Payments, InstructorEarnings
Learning Tracking and Assessments	Per-lesson progress, quizzes and final exams, questions and options, attempts and the answers given.	LessonProgress, Quizzes, Questions, QuestionOptions, QuizAttempts, StudentAnswers
Reviews and Certificates	The single optional review and certificate that an enrollment can produce.	CourseReviews, Certificates

2.7 Relationships

Cardinality is Parent end : Child end. UserRoles and StudentPreferences resolve the two many-to-many relationships. Every row is one foreign key (32 in total). Delete rules are assumed: cascade only for dependent structure rows; history-bearing tables use NO ACTION (Quizzes.SectionId also uses NO ACTION to avoid multiple cascade paths in SQL Server).

#	Parent	Child	Cardinality	Foreign key	On delete	Description
1	**Users**	**UserRoles**	1 : 1..N	UserRoles.UserId	CASCADE	A user holds one or more roles.
2	**Roles**	**UserRoles**	1 : 0..N	UserRoles.RoleId	CASCADE	A role can be assigned to many users.
3	**Users**	**StudentProfiles**	1 : 0..1	StudentProfiles.UserId	CASCADE	A user may have one student profile.
4	**Users**	**InstructorProfiles**	1 : 0..1	InstructorProfiles.UserId	CASCADE	A user may have one instructor profile.
5	**StudentProfiles**	**StudentPreferences**	1 : 0..N	StudentPreferences.StudentId	CASCADE	A student can select many preferred categories.
6	**Categories**	**StudentPreferences**	1 : 0..N	StudentPreferences.CategoryId	CASCADE	A category can be preferred by many students.
7	**Categories**	**Subcategories**	1 : 0..N	Subcategories.CategoryId	NO ACTION	A category contains many subcategories.
8	**Subcategories**	**Courses**	1 : 0..N	Courses.SubcategoryId	NO ACTION	A subcategory classifies many courses.
9	**InstructorProfiles**	**Courses**	1 : 0..N	Courses.InstructorId	NO ACTION	An instructor (user with an instructor profile) owns many courses.
10	**Courses**	**CourseLearningOutcomes**	1 : 0..N	CourseLearningOutcomes.CourseId	CASCADE	A course lists many learning outcomes.
11	**Courses**	**CourseSections**	1 : 0..N	CourseSections.CourseId	CASCADE	A course is organized into sections.
12	**CourseSections**	**Lessons**	1 : 0..N	Lessons.SectionId	CASCADE	A section contains many lessons.
13	**StudentProfiles**	**Enrollments**	1 : 0..N	Enrollments.StudentId	NO ACTION	A student (user with a student profile) can enroll in many courses.
14	**Courses**	**Enrollments**	1 : 0..N	Enrollments.CourseId	NO ACTION	A course can have many enrollments.
15	**Enrollments**	**Payments**	1 : 0..N	Payments.EnrollmentId	NO ACTION	An enrollment can have payment records (e.g., retries, refunds).
16	**Enrollments**	**CourseReviews**	1 : 0..1	CourseReviews.EnrollmentId	NO ACTION	An enrollment can have at most one review.
17	**Enrollments**	**Certificates**	1 : 0..1	Certificates.EnrollmentId	NO ACTION	An enrollment can have at most one certificate.
18	**Enrollments**	**LessonProgress**	1 : 0..N	LessonProgress.EnrollmentId	NO ACTION	Progress is tracked per enrollment; (EnrollmentId, LessonId) is unique.
19	**Lessons**	**LessonProgress**	1 : 0..N	LessonProgress.LessonId	NO ACTION	A lesson has progress rows for many enrollments.
20	**Enrollments**	**QuizAttempts**	1 : 0..N	QuizAttempts.EnrollmentId	NO ACTION	An enrollment can have many quiz attempts.
21	**Quizzes**	**QuizAttempts**	1 : 0..N	QuizAttempts.QuizId	NO ACTION	A quiz can be attempted many times.
22	**Courses**	**Quizzes**	1 : 0..N	Quizzes.CourseId	CASCADE	A course has section quizzes and one final exam.
23	**CourseSections**	**Quizzes**	0..1 : 0..N	Quizzes.SectionId	NO ACTION	A quiz may belong to a section; NULL = course-level quiz or final exam.
24	**Quizzes**	**Questions**	1 : 1..N	Questions.QuizId	CASCADE	A quiz contains one or more questions.
25	**Questions**	**QuestionOptions**	1 : 1..N	QuestionOptions.QuestionId	CASCADE	A question has multiple answer options.
26	**QuizAttempts**	**StudentAnswers**	1 : 0..N	StudentAnswers.AttemptId	NO ACTION	An attempt stores its answers.
27	**Questions**	**StudentAnswers**	1 : 0..N	StudentAnswers.QuestionId	NO ACTION	A question is answered in many attempts.
28	**QuestionOptions**	**StudentAnswers**	0..1 : 0..N	StudentAnswers.SelectedOptionId	NO ACTION	Each answer references a question and optionally a selected option.
29	**Payments**	**InstructorEarnings**	1 : 0..N	InstructorEarnings.PaymentId	NO ACTION	A payment generates instructor earning records.

#	Parent	Child	Cardinality	Foreign key	On delete	Description
30	**InstructorProfiles**	**InstructorEarnings**	1 : 0..N	InstructorEarnings.InstructorId	NO ACTION	An instructor receives many earning records.
31	**Courses**	**InstructorEarnings**	1 : 0..N	InstructorEarnings.CourseId	NO ACTION	Earnings are attributed to a course.
32	**InstructorEarnings**	**InstructorEarnings**	0..1 : 0..N	InstructorEarnings.RelatedEarningId	NO ACTION	Optional self-reference: a reversal entry points to the earning it reverses.